

LiPON — phosphate-oxynitride applicability limit

A chemically constrained network provides the most stringent pretrained-potential test

1 Literature

Lacivita et al. (2018)
Neutron / IR + AIMD
Seth et al. (2025): NequIP
Bates et al. (1996): experiment

2 Model & preparation

$\text{Li}_{47}\text{P}_{16}\text{O}_{56}\text{N}_5$; 124 atoms
Preparation B; NEP89 / GPUMD
2000 K hold: 10 ps
Quench $\rightarrow$ 250 K: 7 ps
250 K hold + release: 20 + 20 ps

3 Transport MD

T: 600 / 900 / 1200 / 1500 K
Ramp: 10 ps; NPT: 50 ps
NVT: 300 ps
600 K primary run: 600 ps
dt: 0.5 fs

4 Analysis

MSD $\rightarrow$ $D(T)$, σ_{NE} and E_a
P-O / P-N / Li RDF
N topology and contacts
NequIP / experiment comparison

5 Conclusion

Thermal trend reproduced
D strongly overestimated
Transferability limit exposed